Data Analytics Project Plan

YouTube Comment Analytics: Doug DeMuro vs. Seth's Bike Hacks

1. Introduction

This project applies the data science pipeline to an audience analytics problem in digital media and the creator economy by comparing how viewers engage with and feel about two YouTube influencers: Doug DeMuro (car reviews) and Berm Peak (Seth's Bike Hacks) (bicycle content). Brands, sponsors, and creators need more than view counts to assess channel health. They need to know whether audience sentiment is trending positively, how it compares across channels, and whether engagement patterns, volume, timing, and emoji use support that sentiment.

The plan below walks through each stage of the pipeline, from data source and ingestion to engineering, analytics computation, and visualization, and describes the work, estimates effort, and shows the anticipated output for each stage. Because this plan was developed alongside a working prototype, the “anticipated results” shown are the pipeline's actual outputs, which both validate the plan and show what a reviewer should expect to see.

2. Data Source

Data is pulled directly from the YouTube Data API v3 (search, channels, and commentThreads endpoints). For each of the two channels, the plan targets the 5 most-viewed videos (10 videos total) and collects up to 1,000 top-level comments per video, yielding roughly 10,000 raw comment records.

Fields captured

video_id, comment_id, author, text, like_count, published_at (UTC), influencer (channel label), video_title.

Sample of raw data

Influencer	Author	Comment (truncated)	Likes	Published
Doug DeMuro	@DougDeMuro	Welcome to Dougcember!	7,109	2019-12-03
Doug DeMuro	@MastaFRO1	"...it's got a 72 Doug score..."	245	2019-12-03
Seth's Bike Hacks	@sbh_fan22	This build is insane, love the detail	312	2021-06-14

Table 1. Three representative rows from the raw comment pull (10,000 rows, 8 columns total).

3. Pipeline Stage: Data Ingestion

Work that needs to be completed

- Register a YouTube Data API key and build a quota-aware API client.
- Resolve each channel's handle to a channel ID, then pull its top videos by view count.
- For each of the 10 selected videos, paginate commentThreads.list to collect top-level comments, handling pagination tokens, quota limits, and transient errors with backoff/retry.

Flatten the nested API JSON into a tabular structure and save it to raw_comments_combined.csv.

Estimated effort: 8–10 hours (API setup ~2h, channel/video resolution ~1h, paginated comment collection & error handling ~4–5h, flatten/export/validation ~2h)

Anticipated result

A single raw CSV of ~10,000 rows × 8 columns, structured as shown in Table 1 above.

4. Pipeline Stage: Data Engineering

Work that needs to be completed

- Remove empty or whitespace-only comments.
- Remove duplicate comments (matching text + author).
- Detect and remove non-English comments using language detection, with safe fallback handling for text too short to classify.
- Remove outdated comments posted before a fixed cutoff date, so the analysis reflects a consistent, recent time window.
- Re-validate row/column integrity and export the result to cleaned_comments.csv.

Estimated effort: 6–8 hours (empty/duplicate removal ~1h, language filtering & tuning ~2h, date-cutoff filtering ~1h, validation/export/QA ~2–3h)

Anticipated result - cleaning funnel

Stage	Rows remaining
Raw comments pulled	10,000
After de-duplication + non-English removal	9,352
After outdated-comment (pre-cutoff) removal — final cleaned set	5,895

Table 2. Row counts at each cleaning checkpoint (actual pipeline output).

5. Pipeline Stage: Analytics Computation

Work to be done

- Basic descriptive statistics: total/unique comment counts, unique reviewer counts, and word-count distribution overall and by influencer.
- Sentiment analysis: score each comment with NLTK VADER (compound score), classify positive/neutral/negative at ±0.05 thresholds, and aggregate by influencer.

- Time-series analysis: bucket comments by month and track comment volume and average sentiment over time, per influencer.
- Emoji sentiment analysis: extract emoji from each comment, map each to a sentiment label, classify by majority vote, and cross-tabulate against the text-based sentiment label to check agreement between the two signals.

Estimated effort: 14–16 hours (basic stats ~2h, sentiment scoring & threshold tuning ~4h, time-series grouping & trend validation ~4h, emoji extraction/mapping/cross-tab ~4–5h)

Anticipated result - sentiment breakdown

Influencer	Positive	Neutral	Negative
Doug DeMuro	51.5%	30.9%	17.6%
Seth's Bike Hacks	45.7%	30.0%	24.3%

Table 3. Share of cleaned comments by sentiment class and influencer (actual pipeline output).

6. Pipeline Stage: Visualization

Work that needs to be completed

Seven chart types, each targeting one analytic question, rendered with a consistent color scheme (Doug DeMuro = blue, Seth's Bike Hacks = orange) so the two channels stay visually distinguishable across every figure:

- Bar chart - total comments by influencer
- Bar chart - comments by individual video
- Pie chart - overall sentiment breakdown
- Boxplot - comment word-count distribution by influencer
- Time series - monthly comment volume and sentiment trend
- Scatter plot - comment length vs. sentiment score

- Word clouds - most frequent terms by influencer

Estimated effort: 10–12 hours (≈1–1.5 h per chart for 7 charts ≈ 8–10 h, plus a cross-chart styling/consistency pass ~2 h)

Anticipated results

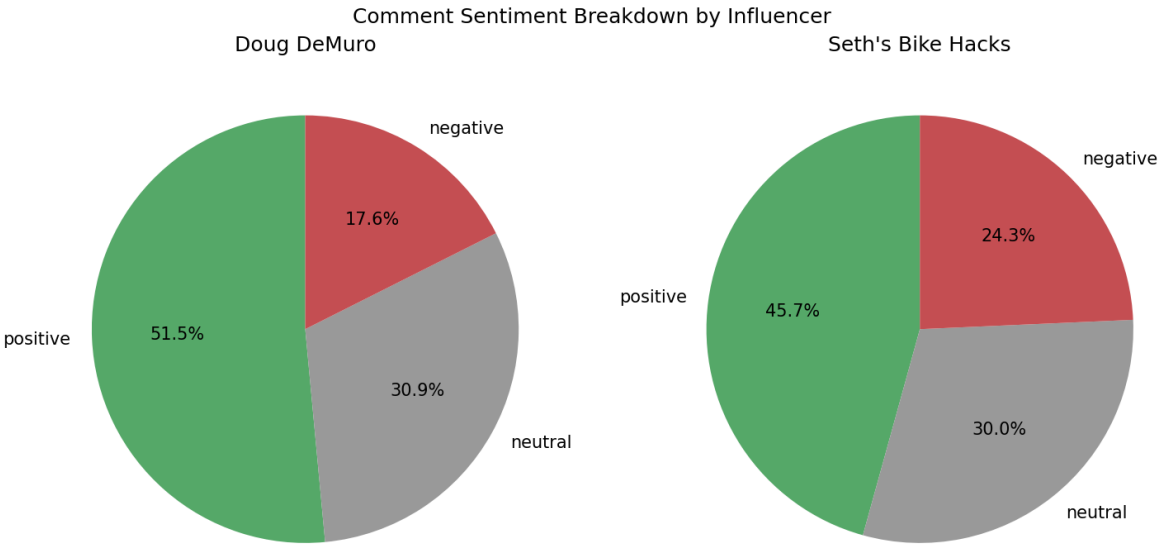


Figure 1. Overall sentiment breakdown (pie chart), actual pipeline output.

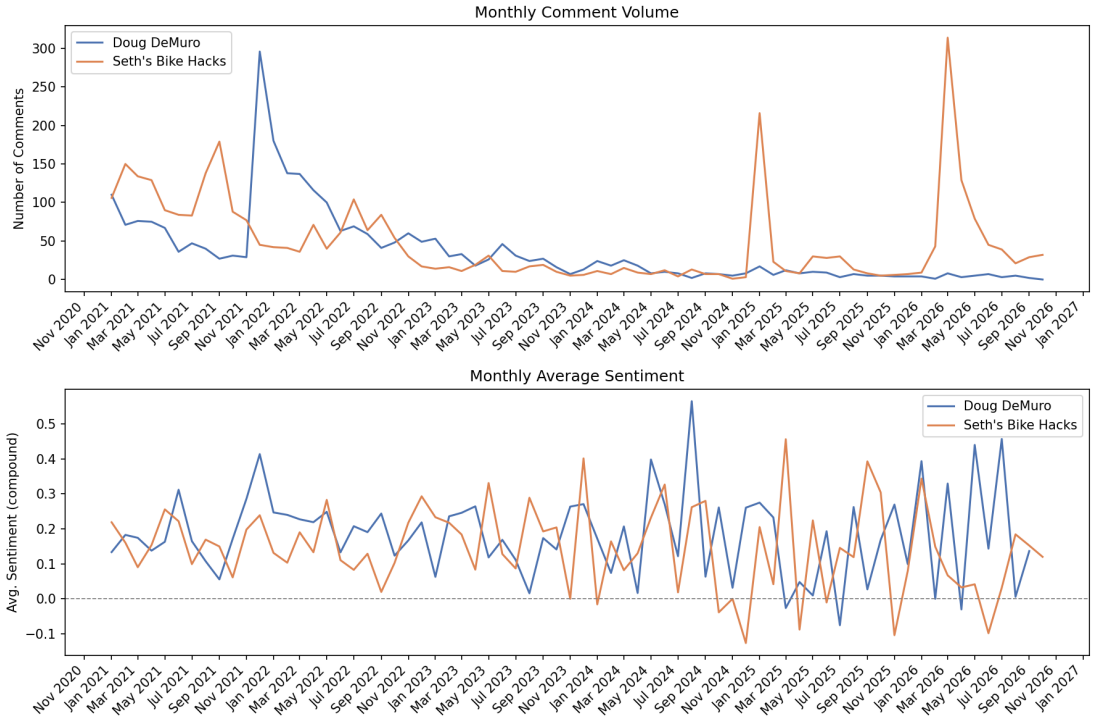


Figure 2. Monthly comment volume and sentiment trend (time series), actual pipeline output.

7. Deliverable Stage

What will be delivered

- A written report (project methodology, findings, and limitations).
- A slide deck sized for a 15-minute presentation.
- A documented, interactive Python program that reproduces every statistic and chart from the raw/cleaned CSVs, including a menu-driven demo mode.
- The raw and cleaned comment datasets (CSV).

Estimated effort: 10–12 hours (report writing ~5h, slide design ~3h, program packaging/testing ~2–3h)

Total estimated project effort: approximately 48–58 hours across all stages above